AgroBot

- Universal AI-Based Agricultural Assistant

Presenting by

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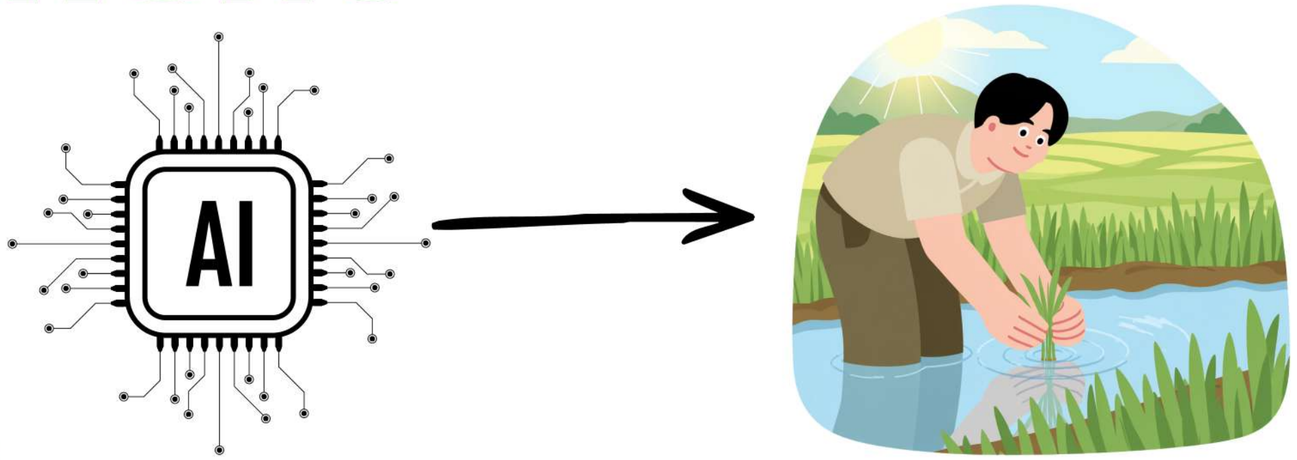
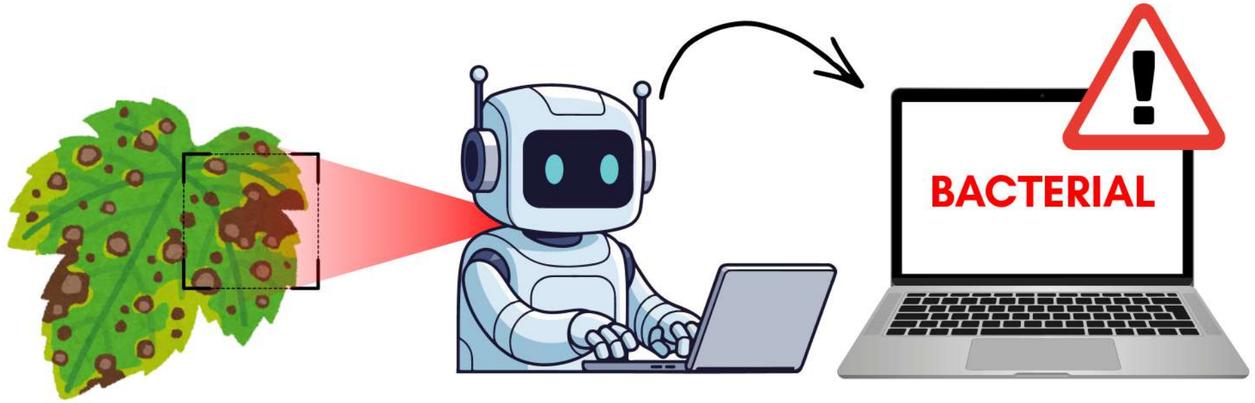
Project Statement

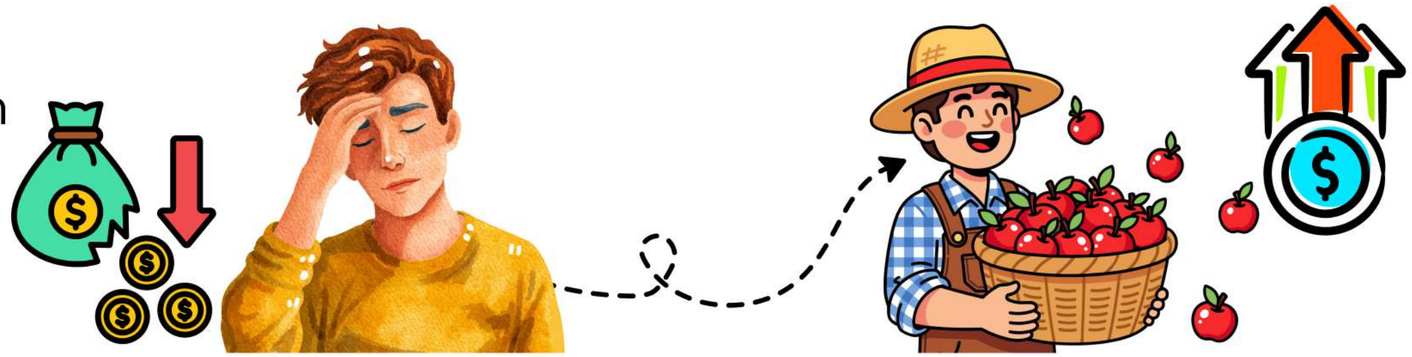
- **AI-based Agricultural Assistant (AgroBot)**– Supports smart decision making for farmers





- Crop recommendation using soil & climate data
- Plant disease detection (Image + Deep Learning)
- Fertilizer suggestions for better productivity
- Weather based farming planning

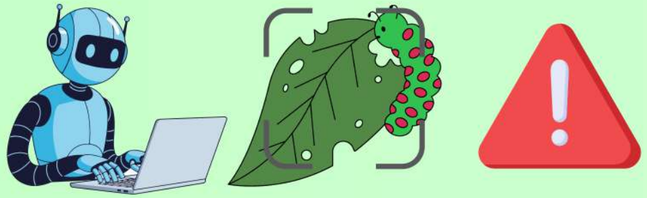
- Easy to use interface for farmers
- All services in one integrated platform
- Improve yield & reduce crop loss
- Promote sustainable farming



Project Outcome

Smart Crop Selection

Accurate crop recommendation based on soil & climate data



AI-Based Disease Detection

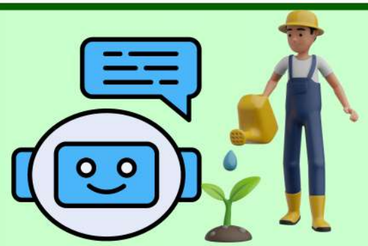
Identifies plant diseases using image analysis

Improved Farming Decisions

Provides intelligent suggestions for better productivity

Real-Time Weather Awareness

Helps plan irrigation and harvesting effectively



Interactive Farmer Support

Real-time chat system for user interaction

Increased Efficiency & Yield

Reduces losses and improves overall crop output

Technology Stack

CV (Computer Vision)

Disease & health analysis, results sent to Chatbot for explanation

Gemini API(Primary AI Engine)

Gemini API

Multimodal Text, Image Processing, translation &utility service API

NLP (Natural Language Processing)

Interprets user intent and generates conversion, managed by Chatbot logic

Core Backend & Framework

Python Programming language

Flask Web Framework

Flask- Socket.io
Realtime-ChatBot

AgroBot System

Data Management & Storage

SQLite SQLite Database

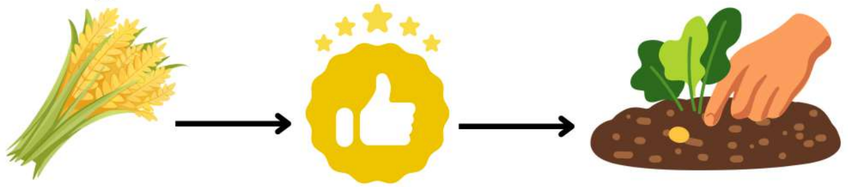
SQLAlchemy
ORM for SQLite

External Integeration(APIs)

OpenWeatherMap
Weather Data Service

Modules to be implemented

1. Crop Recommendation Module



- Analyzes soil nutrients and environmental conditions
- Suggests the most suitable crops for better yield

2. Disease Detection Module

- Accepts crop images from users
- Identifies plant diseases using AI

3. Fertilizer Recommendation Module

- Provides fertilizer suggestions based on soil and crop
- Helps improve productivity and maintain soil health

4. Weather Monitoring Module

- Fetches real-time weather data using API
- Assists in planning irrigation and harvesting

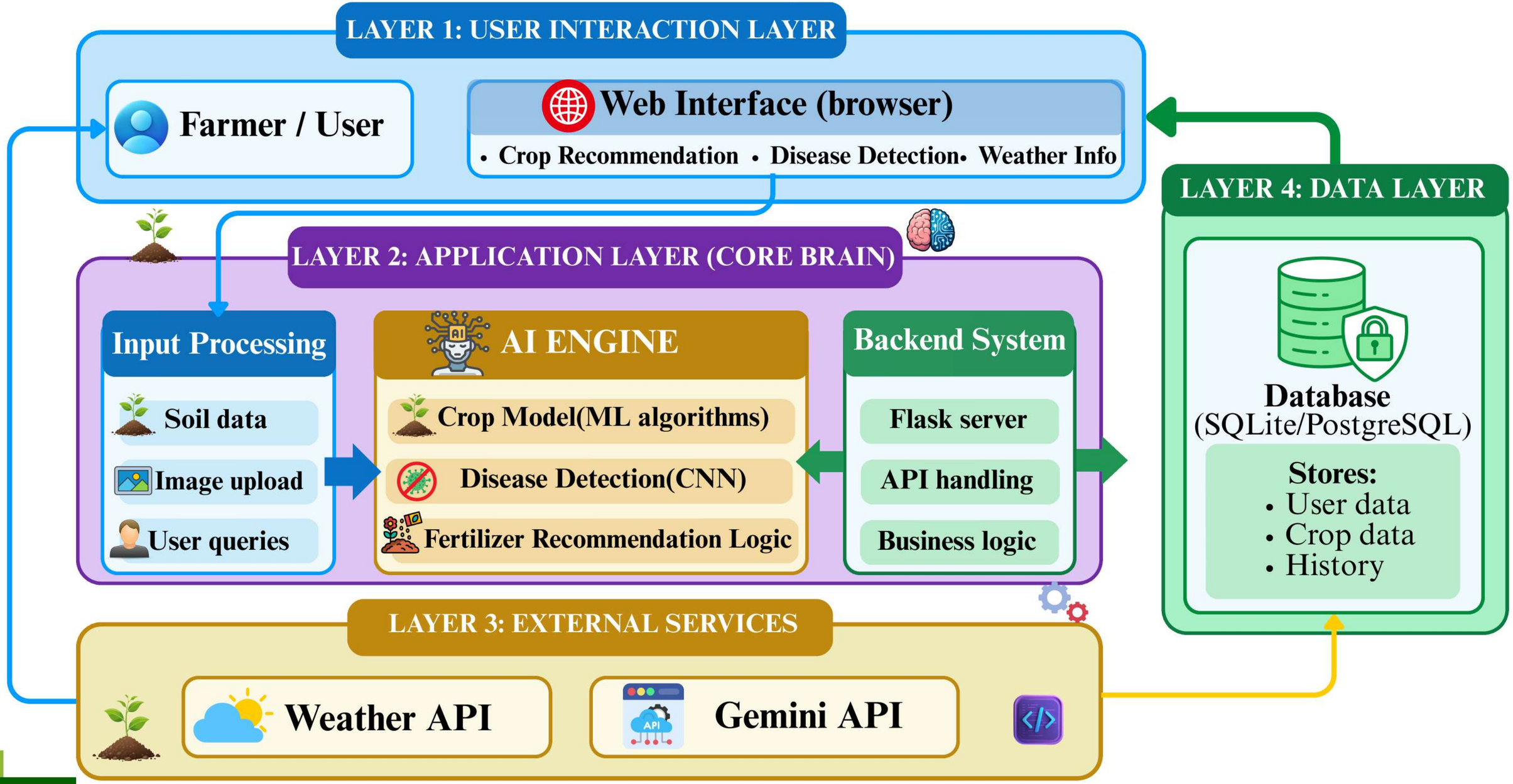
5. Chatbot Assistance Module

- Enables natural language interaction using Gemini API
- Provides real-time responses to farmer queries

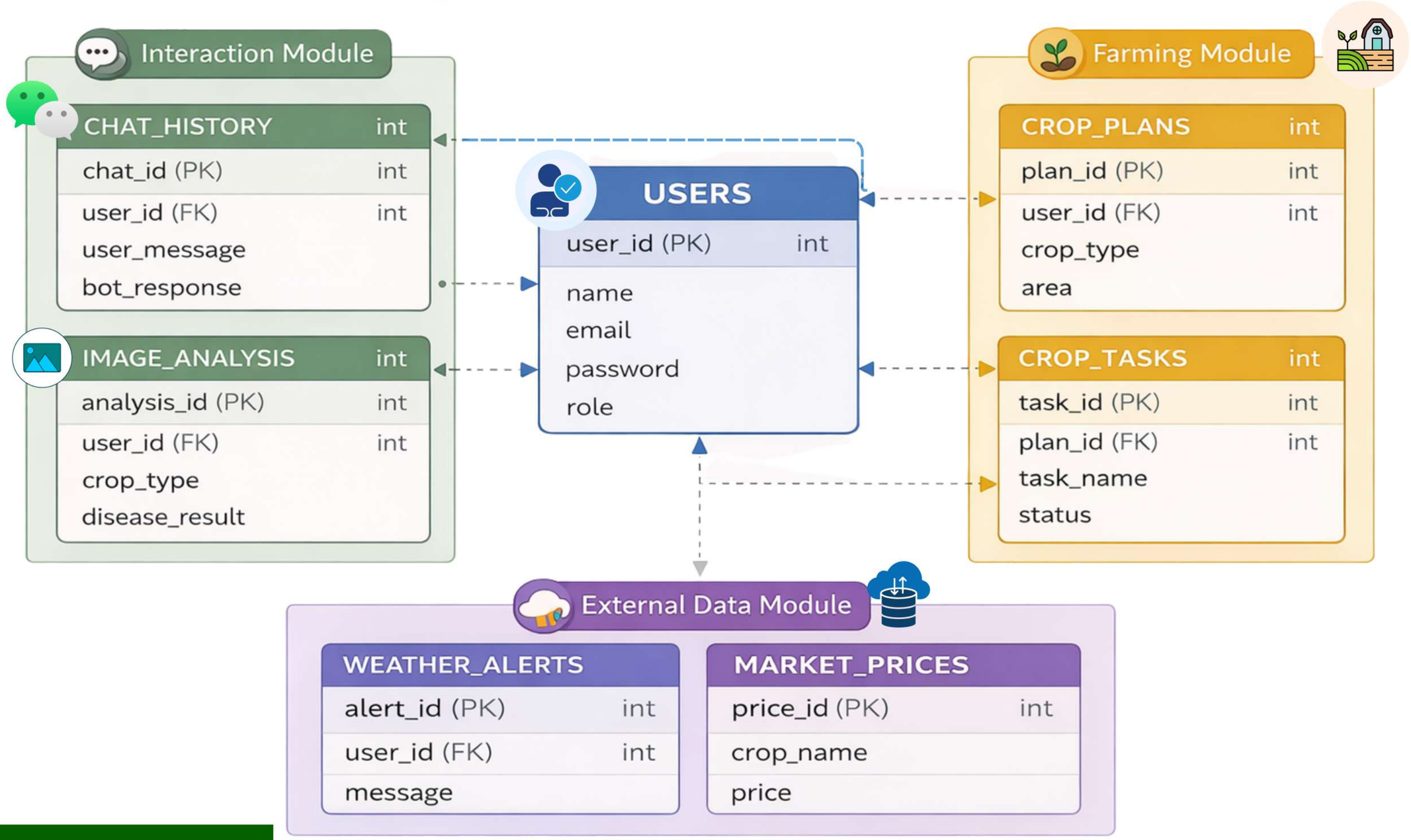
6. Data Management Module

- Stores user data, crop plans, and history
- Manages database operations using SQLite & SQLAlchemy

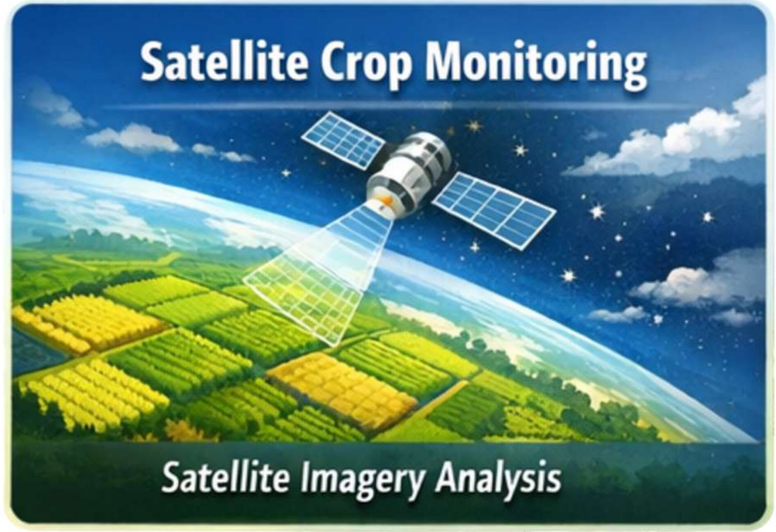
Architecture Diagram



Database Schema



Feature Enhancement



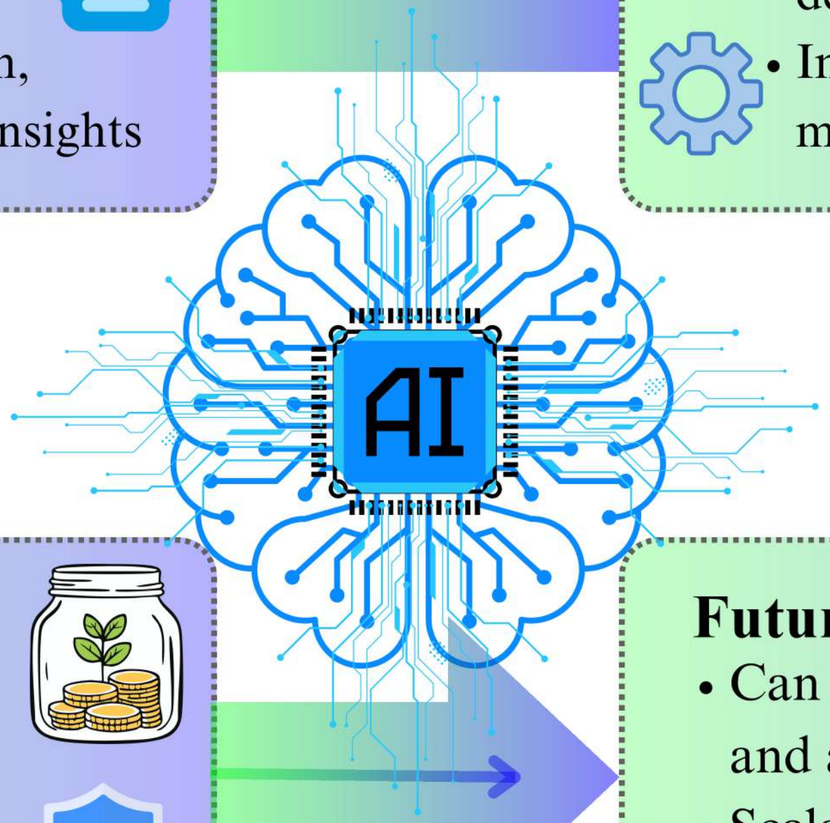
Conclusion

AgroBot System Overview:

- Provides an Al-based solution for smart farming
- Integrates crop recommendation, disease detection, and weather insights

Technology Impact:

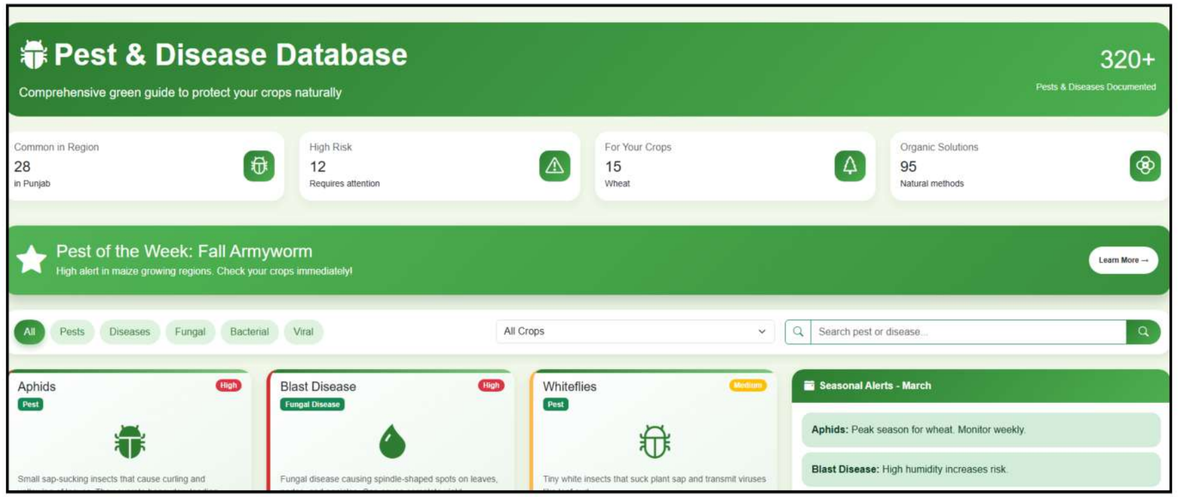
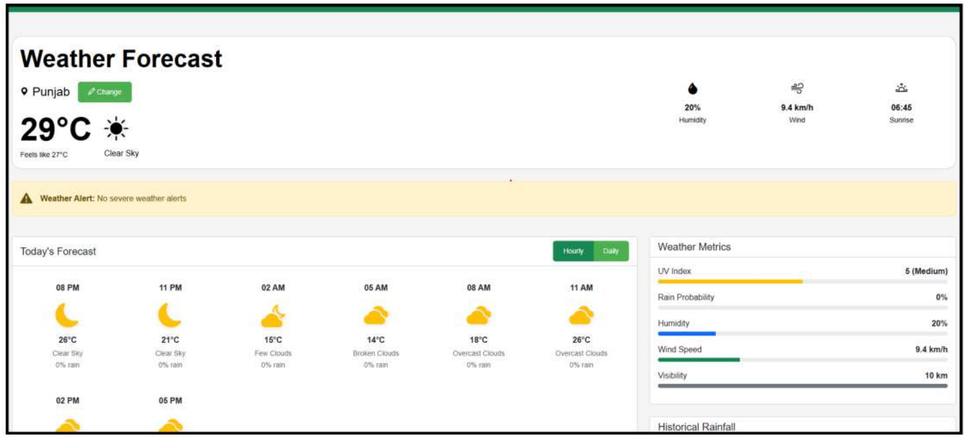
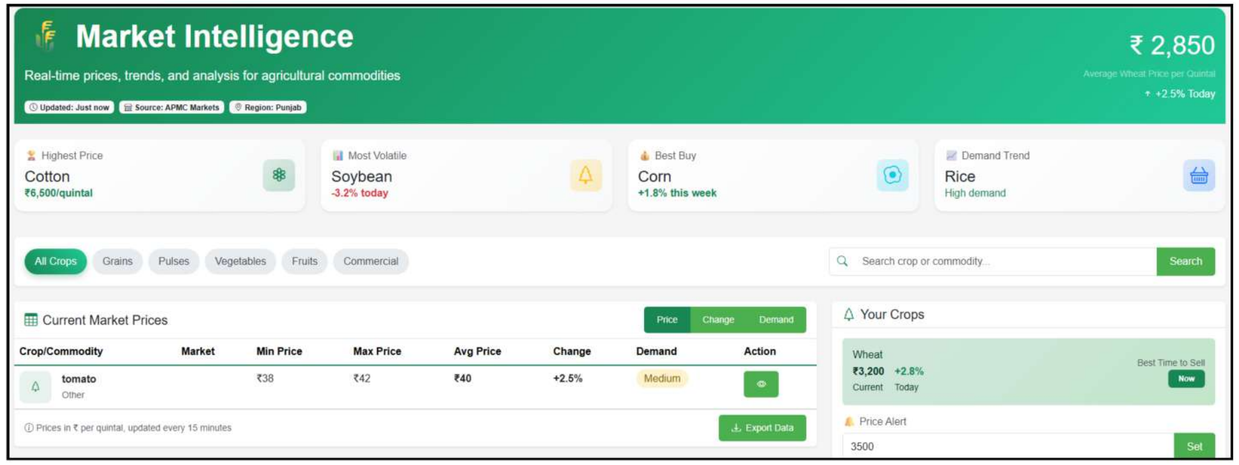
- Uses Al and real-time data for better decision-making
- Improves efficiency and reduces manual effort



Benefits to Farmers:

- Enhances productivity and crop yield;
- Reduces risks and farming losses

Future Scope:

- Can be enhanced with loT, satellite, and advanced analytics;
- Scalable for real-world agricultural applications

Sample Output